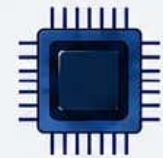
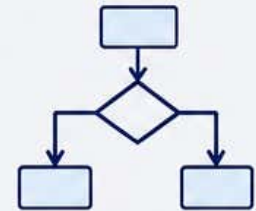




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Syllabus



Major Course (Elective-2 Laboratory): Laboratory on Artificial Intelligence

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	112227	Laboratory on Artificial Intelligence	2	60	4Hrs	50

Course Objectives:	1. To learn the syntax and semantics of the Python programming language. 2. Write and Debug Code: Develop executable Python scripts using control structures, loops, and custom functions. 3. Handle Data: Utilize AI-specific Python libraries (e.g., NumPy and pandas) for data cleaning, pre-processing, and manipulation. 4. Build AI Models: Use frameworks like PyTorch or TensorFlow to construct, train, and evaluate basic machine learning models. 5. Problem Analysis: Formulate, design, and interpret data-driven solutions and predictive models for real-world scenarios.			
Course Outcomes:	On completion of the following syllabus the students will be able to - 1. Apply blind search techniques for AI problem solving. 2. Data Literacy & Engineering: Proficiency in manipulating large datasets using libraries like NumPy for numerical data and Pandas for structured data. 3. Apply mathematical foundations and algorithmic principles to model and solve complex engineering problems. 4. Ability to perform Knowledge Acquisition—gathering domain-specific expertise from human experts and documenting it for a knowledge base. 5. Computer Vision & NLP: Competency in processing digital images and videos, as well as text-based sentiment analysis and tokenization.			
Contents		Workload Allotted	Weightage of Marks Allotted	Incorporation of Pedagogies
	List of Practical: 1. Write a Program to print multiplication table for given no. 2. Write a Program to check whether the given no is prime or not. 3. Write a Program to find factorial of the given no. 4. Write a program to implement List Operations 5. Write a program to Illustrate Different Set Operations. 6. Write a program to implement Simple Chatbot. 7. Write a Program to Implement Breadth First Search. 8. Write a Program to Implement Depth First Search 9. Write a program to implement Hill Climbing Algorithm 10. Write a program to implement A* Algorithm			Use any pedagogical technique suitable for the topic

	11. Write a program to solve Traveling salesman problem 12. Write a program to solve Water jug problem 13. Write a program to implement Regression algorithm. 14. Write a program for chomsky hierarchy grammar stratification.			
References:	Web link to Equivalent MOOC on SWAYAM if relevant: 1. https://www.programiz.com/pythonprogramming/examples/multiplication-table 2. https://www.programiz.com/python-programming/examples/prime-number 3. https://realpython.com/build-a-chatbot-python-chatterbot/ 4. https://www.geeksforgeeks.org/python/python-program-for-breadth-first-search-or-bfs-for-a-graph/			

Assessment Rubric: External Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)				
Practical	Course Code: 112227	Laboratory on Artificial Intelligence	Max Marks: 25	Time: 4 Hrs.
Sr. No.	Exercise			Marks
1.	Single Program Writing, Algorithm, Flowchart, Execution, with Sample inputs and Outputs, and Printout			10
2.	Submitted Record book (Preferably handwritten)			05
3.	External Viva Voce			10

Assessment Rubric: Internal Practical Course

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI PRACTICAL EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Practical	Course Code: 112227	Laboratory on Artificial Intelligence	Max Marks: 25
Sr. No.	Assessment Criteria		Marks
1.	Attendance/Regularity in practical's / Students' overall performance		05
2.	Record book Preparation (Preferably handwritten)		10
3.	Internal Viva Voce		10